

Saniya Patwardhan

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Eligible for Graduate Worker Route (GWR) visa in UK for two years after graduation

EDUCATION

- 2025 – present **Master's Degree (MRes Research at Robot Intelligence Lab, Dyson School of Design Engineering)** ongoing
Imperial College London
India Future Leaders Scholarship 2025
- 2020 – 2024 **B.Tech in Mechanical Engineering (Minor in Robotics)** GPA: 8.75/10
Indian Institute of Technology Gandhinagar
Director's Silver Medal for Outstanding Overall Performance

EXPERIENCE

Robotics Intern, Embodied AI and Machine Vision May 2024 - Present
Netholabs (Part-time)

- Building robotics and computer vision pipelines for an early stage AI-native neurotech startup.

Postgraduate Researcher, Embodied AI Aug 2024 - Jul 2025
Imperial College London — Human-Robot Collaboration through VLA-based Learning

- Fine-tuned a large-scale vision-language-action model for robot manipulation tasks using parameter-efficient training methods.
- Integrating human-intention prediction, and adaptive real-time planning to enhance the naturalness and efficiency of collaboration.

Robotics Researcher, Multi-object Grasping Project Aug 2024 - Jul 2025
IITGN Robotics Lab — Sabarmati Bridge Fellowship

- Developed a learning-augmented grasp planning framework combining kinematic reachability and data-driven prediction for multi-object manipulation.
- Developed machine learning models for grasp planning and real-time decision-making under workspace constraints. Developed machine learning models for grasp planning and real-time decision-making under workspace constraints.

Research Intern, Active Robot Vision Project May 2023 - Jul 2023
University of Washington, Seattle (USA)

- Improved object recognition scores in cluttered scenes with occluded objects by enabling a mobile robot to predict and move to the next-best view using an extremum seeking controller.
- Implemented the pipeline using ROS Kinetic on NVIDIA Jetson AGX Orin with Intel Realsense, incorporating a 3D point-cloud-slicing-based topological descriptor for object recognition.

PROJECTS

Open-Source Robot Learning on SO-101 (LeRobot Platform) Ongoing
Self-initiated Project

- Implementing imitation learning pipelines (behavior cloning, diffusion policies) on real robot hardware using demonstration data for manipulation tasks.
- Iterating on data quality and policy performance by curating demonstration datasets and analyzing failure modes to improve robustness under real-world conditions.

GalPal.ai — Context Aware Local LLM-Powered Wellbeing Assistant Sep 2025 – Dec 2025

- Developed a privacy-preserving, local LLM-powered assistant integrated with an iPad to analyze handwritten journal entries, infer user mood, and store structured insights in a SQLite database for Retrieval-Augmented Generation journal insights.
- Implemented a context-aware outfit recommendation pipeline that combines real-time weather data, inferred emotional state from journal analysis, and a self-curated, annotated personal wardrobe dataset.

Autonomous Campus Shuttle

Aug 2023 – May 2024

Indian Institute of Technology Gandhinagar

[\[Video Demo\]](#) — [\[News Article\]](#)

- Built a perception-driven navigation system combining vision and sensor fusion for autonomous decision-making in real-world campus environments.
- Designed and actuated control systems for braking, acceleration, and steering, integrating a closed-loop feedback system using Pixhawk, RGB-D cameras, and LiDAR sensors for autonomous and remote operation.

TECHNICAL SKILLS

Core Areas:	Robot Manipulation, Imitation Learning, Active Perception, Control Systems, Sensor Integration
Programming & Frameworks:	Python, C++, MATLAB, ROS 1 & 2, Arduino, Bash, CMake, OpenCV, Pytorch, Tensorflow
Tools:	Mujoco, Gazebo, RViz, MoveIt, Simulink, Docker, Git, Jupyter
Prototyping:	CAD Modelling, 3D Printing, Laser Cutting, CNC Machining, CNC Machining, Lathe, Welding

ACHIEVEMENTS

- **India Future Leaders Scholarship, Imperial College London:** Awarded GBP 10,000 towards tuition fees; one of only 10 recipients selected from the entire Indian student cohort at Imperial.
- **Student Travel Grant for ICRA 2024:** Received USD 1,000 from the Mobile Manipulation TC and Robot Learning TC to attend ICRA 2024 and participate in the 2nd Workshop on Mobile Manipulation and Embodied Intelligence.
- **Sabarmati Bridge Fellowship, IIT Gandhinagar:** Awarded a year-long fellowship to conduct full-time research at the IITGN Robotics Lab as a Predoctoral Researcher (2024–25).
- **Director’s Silver Medal:** Honored for outstanding overall performance at the 13th Convocation Ceremony of IIT Gandhinagar.
- **UW Mechanical Engineering Summer Research Grant:** Received USD 5,700 to pursue a 10-week research internship at the University of Washington, Seattle.

PUBLICATIONS

- **S. Patwardhan**, A. Ansari, A. Krishna, H. J. Palanthandalam-Madapusi. “*Grasp Sequence Planning for Multi-object Grasping using Reachability Maps.*” Presented at *Advances in Robotics 2025*, 7th International Conference of the Robotics Society. [In print]
- V. K. Jonnalagadda, C. K. Mullapudi, **S. Patwardhan**, E. U. Samani, and A. G. Banerjee. “*Extremum-Seeking Active Object Recognition in Clutter Using Topological Descriptors.*” Presented at *IEEE International Conference on Robotics and Automation (ICRA 2024)*, PACIFICO Yokohama: 2nd Workshop on Mobile Manipulation and Embodied Intelligence. [\[Link\]](#)
- A. Dan, **S. Patwardhan**, S. K. Saha, and K. RamaKrishna. “*A Novel Control Strategy for Stance Stability of a Quadruped Robot against External Disturbance.*” Presented at *Advances in Robotics 2023*, 6th International Conference of the Robotics Society. [\[Link\]](#)